Chapter 1.1

Introduction to Mobile Application Development

Objectives

-Mobile operating systems and ecosystem

-Development tools

-Mobile devices

-Mobile application software architecture and framework

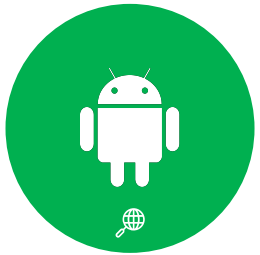
Mobile Operating Systems And Ecosystem

Mobile Operating Systems

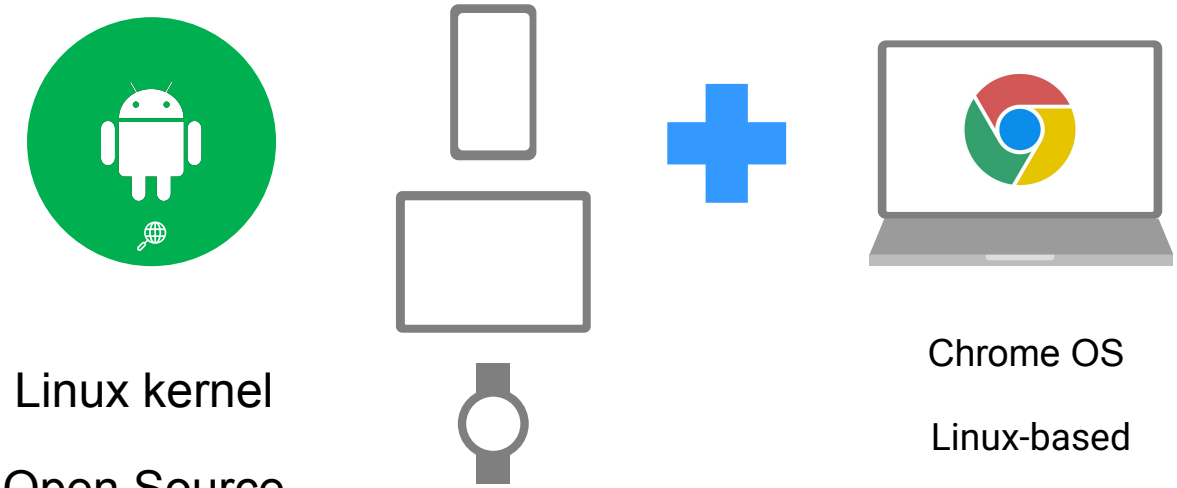
Question?

What is your phone OS?

Mobile OS



Mobile OS - Android



Linux kernel

Open Source

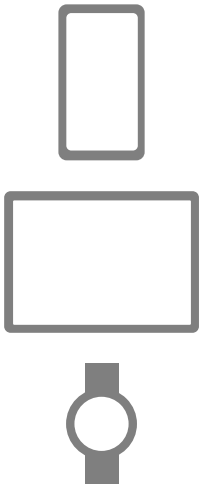
Chrome OS

Linux-based

Open Source

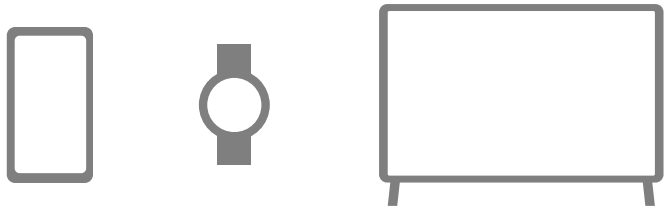
Mobile OS - iOS

Unix kernel



macOS

Mobile OS - Tizen



Linux kernel

Open Source

Mobile OS - KaiOS

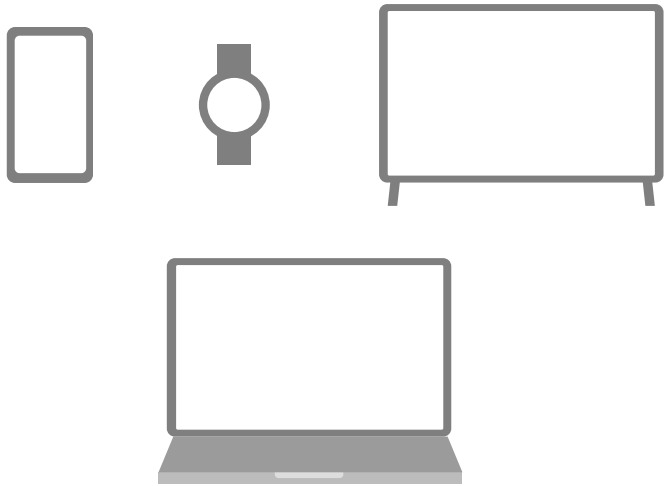
Linux kernel

Open Source

Mobile OS – Harmony OS

Microkernel

Open Source

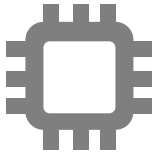


Mobile OS - Fuchsia

Microkernel

Open Source

Module Architecture



ARM-based

Nest Hub

Is Windows CE still available?

Windows CE Mobile Devices

How many types of mobile apps are there?

Types of Mobile Apps

Native

Hybrid

Web
+
Native

Web

Comparison of Apps

Criteria	Native	Hybrid	Mobile Web
Development cost	High	Low	Low
App performance	Fast	Depending on speed of network	Depending on speed of network
Distribution method	App stores	App stores	None
Access to device features	Wide	Limited	Very limited
Code maintenance	Multiple codebase	Single codebase	Single codebase

Question?

In your opinion, which mobile app development method (native, mobile web, and hybrid) is suitable for each the following entities?

a. Bernama, the Malaysian national news-agency
b. Super Mario Bros, a game developed by Sega
c. Setapak Central, a shopping mall

Development Tools

Programming Languages

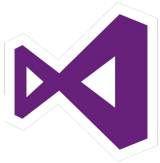
Native	Hybrid	Web
    	Web + Native	  

Development Tools - Native

Android Studio

XCode

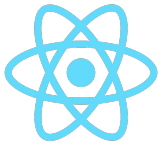
Tizen Studio



Visual Studio

Huawei Quick App

Development Tools – Cross Platform


React Native


Flutter


Xamarin


PhoneGap

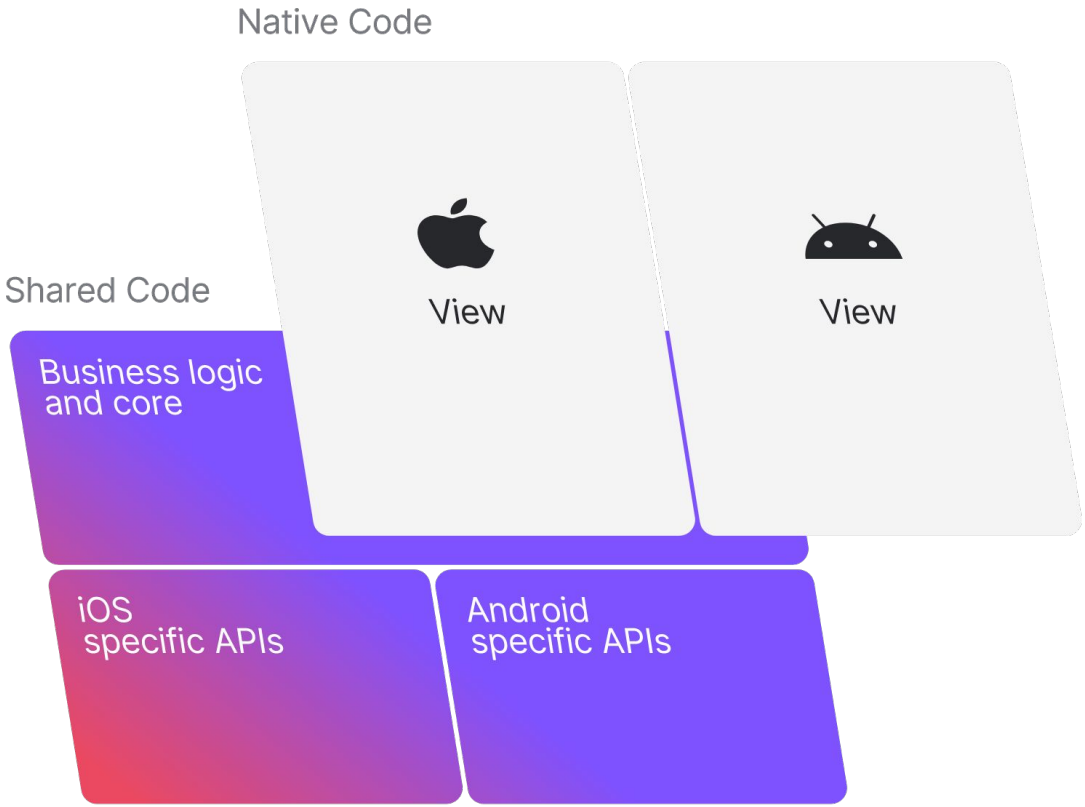

Apache Weex


Native Script


Jasonette


Ionic

Kotlin Multiplatform Mobile (KMM)



https://kotlinlang.org/docs/multiplatform-mobile-getting-started.html

What are the apps developed using KMM?

Real-world success stories

Mobile Devices

Mobile devices

Flagship

Mid Range

Entry

Basic

4610 MHz

1300 MHz

24++ GB RAM / 1TB ROM

4 GB RAM / 64 GB ROM

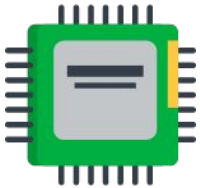
6000 mAh

2150 mAh

5G Network

4G Network

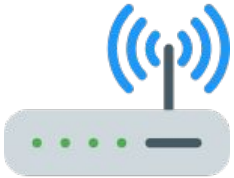
Key Mobile Challenges



Processing Power

Memory and storage

Battery



Network

Beyond Mobile Devices

Beyond Mobile App

Wearable

Smart Home Appliance

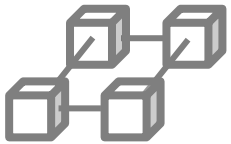
Smart Car Dashboard

Internet of Things
(IOT)

Beyond Mobile App

Augmented Reality
(AR)

Artificial Intelligent
(AI)



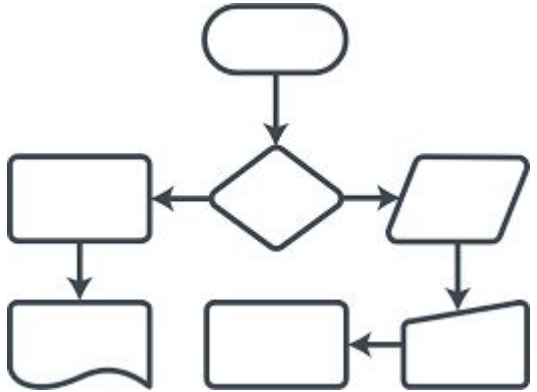
Blockchain

AI Native

AI Native

The focus has shifted from static interfaces to **dynamic**, **context-aware experiences**.

In the past, developers built fixed paths (screens and buttons); today, they build core logic that an AI agent interprets to fulfill user intent.



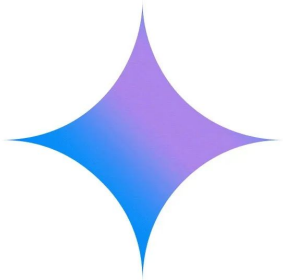
AI Native

Vibe Coding & AI Collaboration

Tools like GitHub Copilot and specialized AI IDEs have turned coding into a high-level architectural task.

Developers often "vibe code" by describing functionality, allowing AI to handle boilerplate, API integration, and even complex logic flows.

GitHub Copilot



Gemini in Android Studio

AI Native

Lower Barrier to Entry

Beginners are now launching sophisticated, profitable apps solo by leveraging generative AI for everything from backend infrastructure to UI assets.

Mobile Application Software Architecture And Framework

Introduction to Flutter

An open-source mobile application development framework created by Google.

It is used to develop applications for Android and iOS, web and desktop (Windows, Mac OS, Linux, etc).

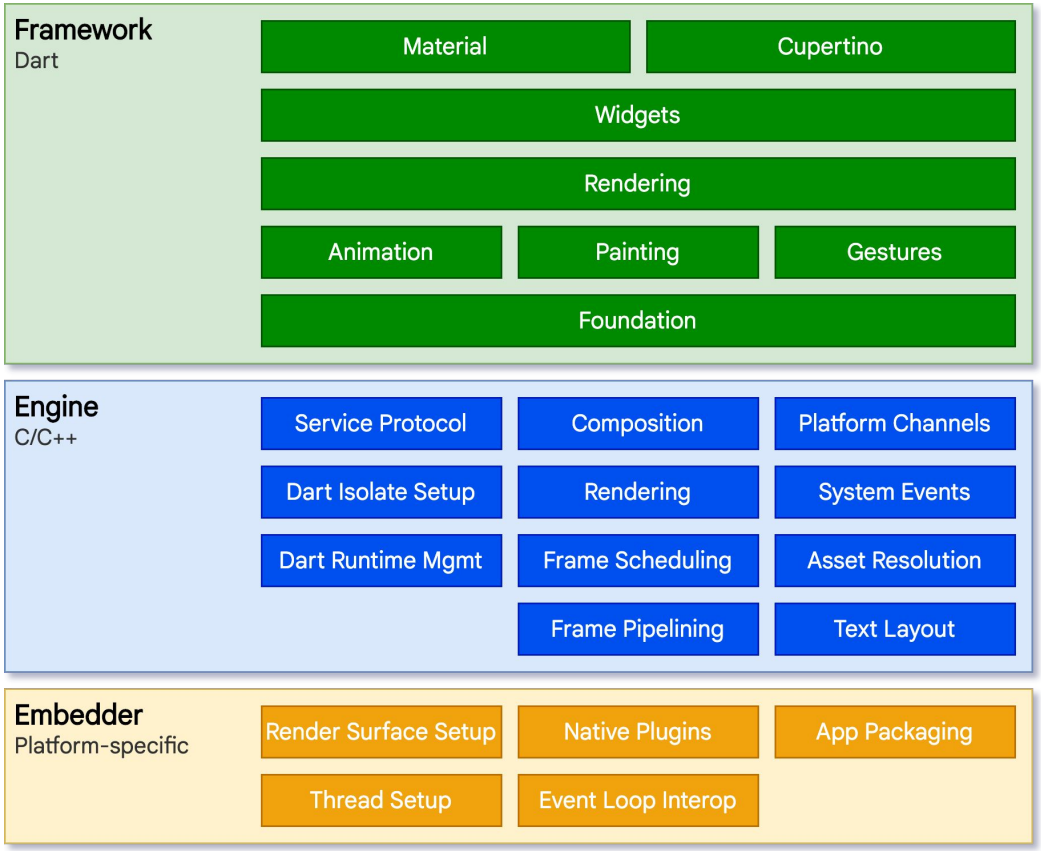
It is an SDK.

Language : DART

Key Features

- Cross-platform development
- Hot reload
- Rich widgets
- Customizable
- High performance
- Open-source

Architectural

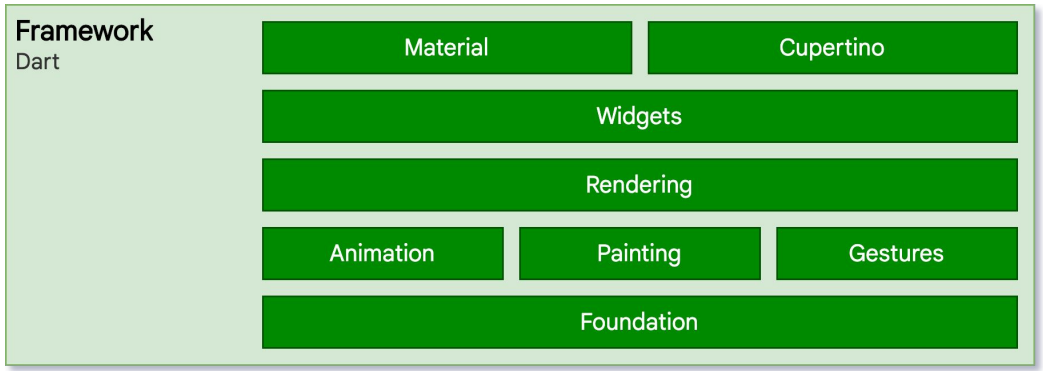


Architectural - Framework

A programming language used to build Flutter apps.

Offers features like strong typing, asynchronous programming, and just-in-time (JIT) compilation for rapid development.

Ahead-of-time (AOT) compilation for production builds to improve performance.

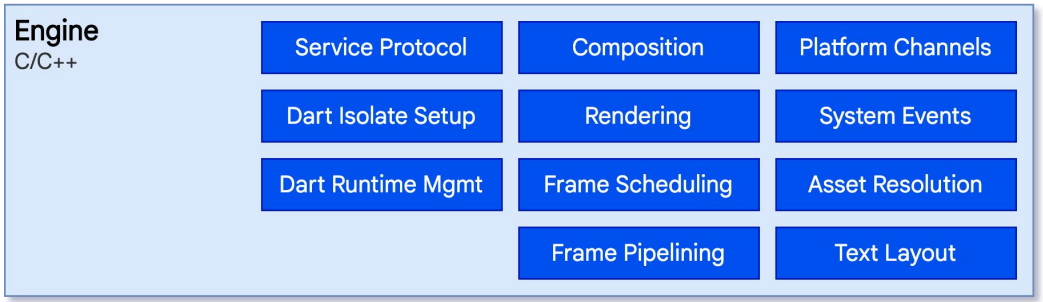


Architectural - Engine

The foundation of Flutter, written mainly in C++.

Handles platform-specific tasks such as rendering, input, and network requests.

Implements the Flutter framework and Dart runtime.



Architectural - Embedder

A platform-specific embedder coordinates with the underlying operating system.

Written in a language that is appropriate for the platform:

- Java and C++ for Android
- Objective-C/Objective-C++ for iOS and macOS
- C++ for Windows and Linux

With embedder, Flutter code can be integrated into an existing application as a module, or the code might be the entire content of the application.

Architectural - Reactive UI

It focuses on data flow and propagation of change.

Building UI that react to changes in data in a declarative and efficient manner.

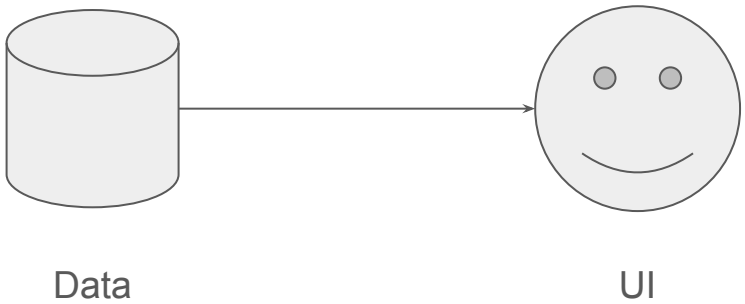
Three key concepts:

1. Data Flow
2. Declarative UI
3. State Management

Architectural - Data Flow

1. Data Flow

Data flows unidirectionally from the data source to the UI. Changes to the data source trigger updates to the UI.



Architectural - Data Flow

1. Data Flow

Reactive UI - a programming paradigm that focuses on data flow and propagation of change.

UI automatically update whenever the underlying data changes.

This leads to more responsive, efficient, and maintainable applications; one-way flow makes it easier to track and debug state changes, as you always know where the data is coming from.

Architectural - Reactive UI

2. Declarative UI

You describe the desired UI state, and Flutter handles the rendering and updates.

This makes it easier to reason about the UI and write less code.

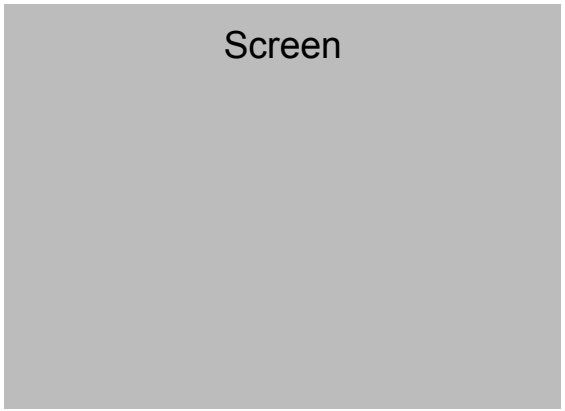
Widget is the fundamental of UI.

```
//Examples of Text widget
Text( 'You have pushed the button this many times:', ),
Text( '$_counter', style: Theme.of(context).textTheme.headline4, ),
```

Architectural - Reactive UI

2. Declarative UI

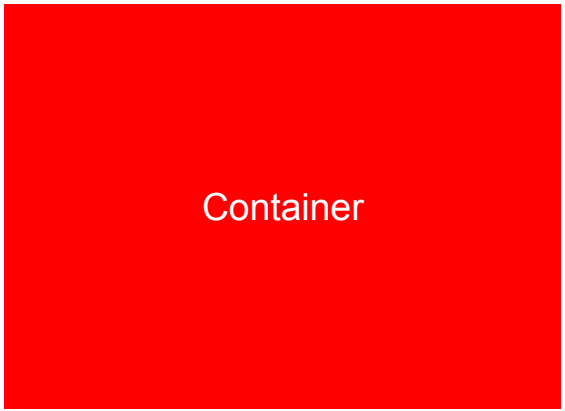
The screen is the parent (the root) of all widgets.



Architectural - Reactive UI

2. Declarative UI

E.g.1: The Container is forced to be exactly the same size as the screen.

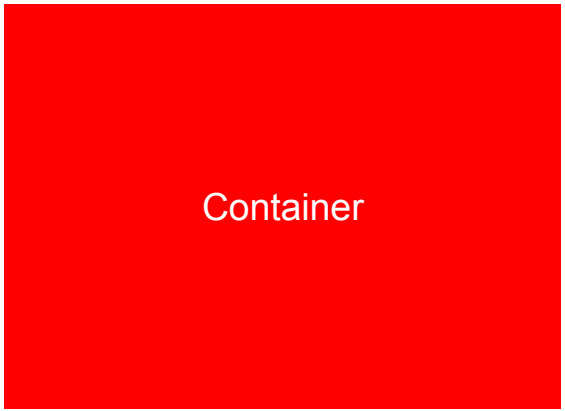



```
Container(color: red)
```

Architectural - Reactive UI

2. Declarative UI

E.g.2: The red Container wants to be 100 × 100, but it can't, because the screen forces it to be exactly the same size as the screen.

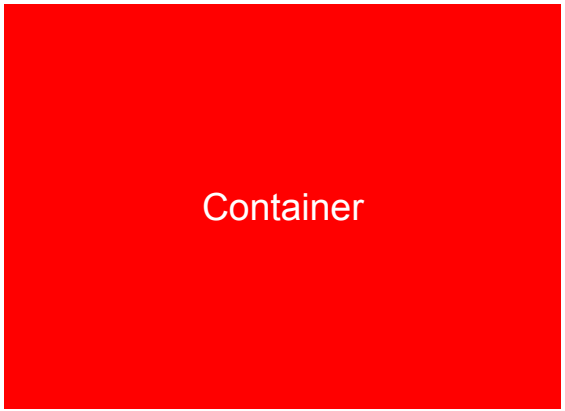



```
Container(
  width: 100,
  height: 100,
  color: red)
```

Architectural - Reactive UI

2. Declarative UI

E.g.3: The screen forces the Center to be exactly the same size as the screen, so the Center fills the screen.

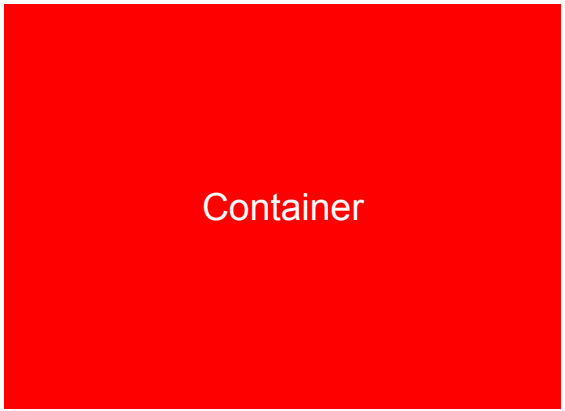



```
Center(
  child: Container(color: red),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.4: The Center tells the Container that it can be any size but not bigger than the screen. The Container wants to be of infinite size, so it just fills the screen.

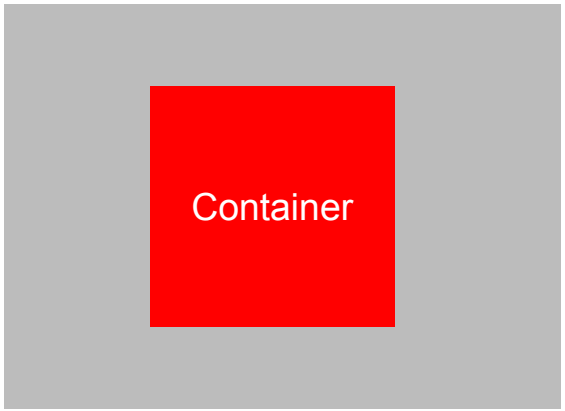



```
Center(
  child: Container(
      width: double.infinity,
      height: double.infinity,
      color: red),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.5: The Center is the same size as the screen. The Center ensures that the Container is not bigger than the screen. So, the Container can be 100 × 100.

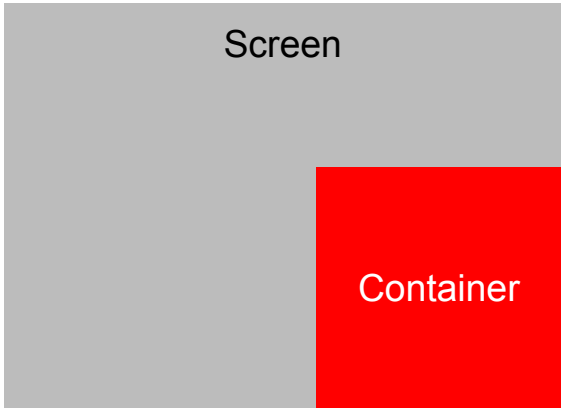



```
Center(
  child: Container(
         width: 100,
         height: 100,
         color: red),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.6: Align tells the Container that it can be any size. It aligns the Container to the bottom-right of the available space.

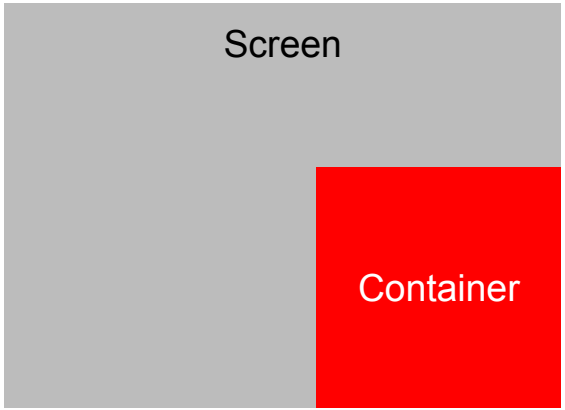



```
Align(
  alignment: Alignment.bottomRight,
  child: Container(
         width: 100,
         height: 100,
         color: red),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.6: Align tells the Container that it can be any size. It aligns the Container to the bottom-right of the available space.

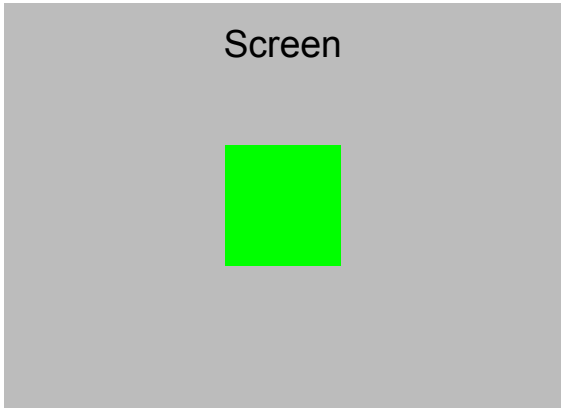



```
Align(
  alignment: Alignment.bottomRight,
  child: Container(
         width: 100,
         height: 100,
         color: red),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.7: The Center tells the red Container that it can as big as the screen. However, the red Container has no size, it decides it wants to be the same size as its child.

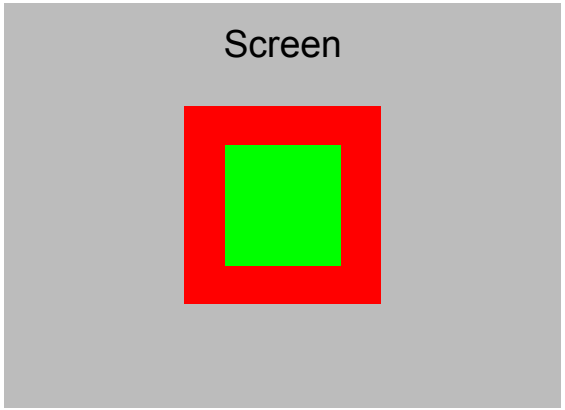



```
Center(
  child: Container(
    color: red,
    child: Container(
            color: green,
            width: 30,
            height: 30),
  ),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.8: The red Container sizes itself to its children's size, but it takes its own padding into consideration. So it is also 30 × 30 plus padding.

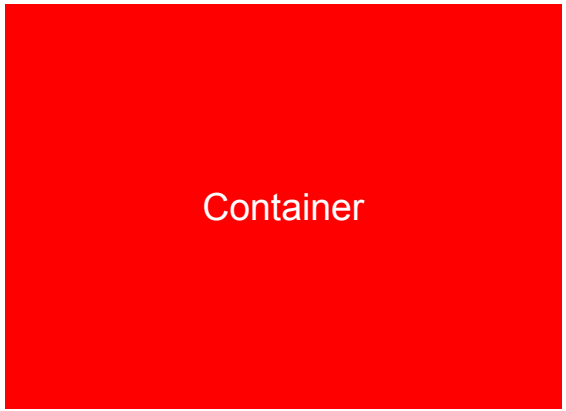



```
Center(
  child: Container(
    padding: const EdgeInsets.all(20),
    color: red,
    child: Container(
           color: green,
           width: 30,
           height: 30
    ),
  ),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.9: The ConstrainedBox is forced to be exactly the same size as the screen, so it tells its child Container to also assume the size of the screen, thus ignoring its constraints parameter.

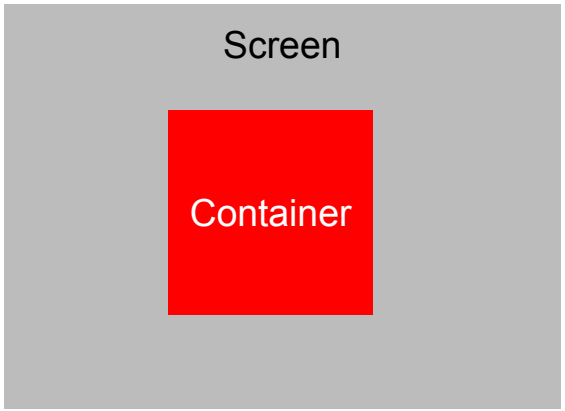



```
ConstrainedBox(
  constraints: const BoxConstraints(
    minWidth: 70,
    minHeight: 70,
    maxWidth: 150,
    maxHeight: 150,
  ),
  child: Container(color: red, width: 10,
                   height: 10),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.10: The Container must be between 70 and 150 pixels. It wants to have 10 pixels, so it ends up having 70 (the minimum).

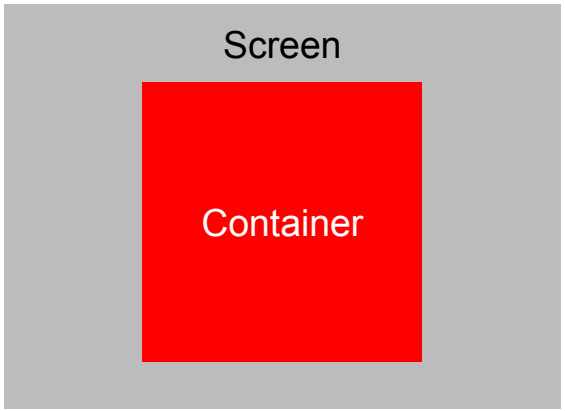



```
Center(
  child: ConstrainedBox(
    constraints: const BoxConstraints(
      minWidth: 70, minHeight: 70,
      maxWidth: 150, maxHeight: 150,
    ),
    child: Container(color: red, width: 10,
                     height: 10),
  ),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.11: The Container must be between 70 and 150 pixels. It wants to have 1000 pixels, so it ends up having 150 (the maximum).

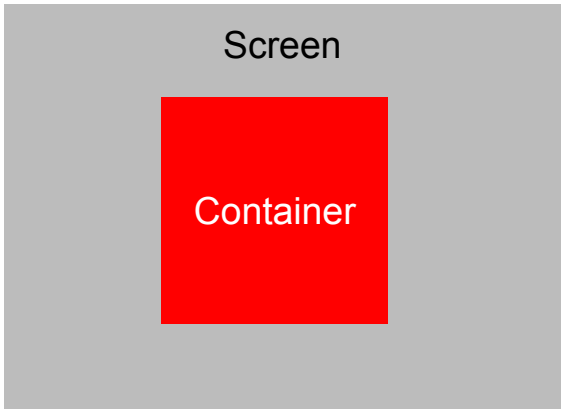



```
Center(
  child: ConstrainedBox(
    constraints: const BoxConstraints(
      minWidth: 70, minHeight: 70,
      maxWidth: 150, maxHeight: 150,
    ),
    child: Container(color: red, width: 1000,
                     height: 1000),
  ),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.12: The Container must be between 70 and 150 pixels. It wants to have 100 pixels, and that's the size it has, since that's between 70 and 150.

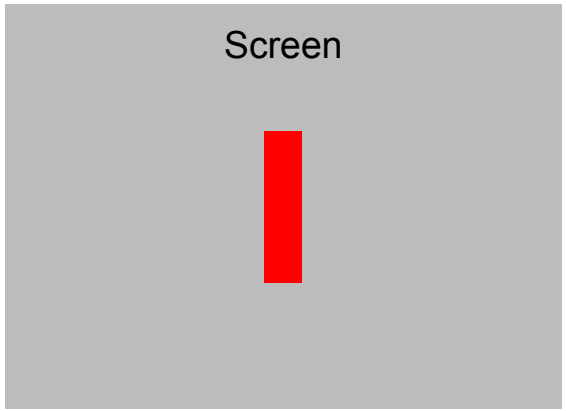



```
Center(
  child: ConstrainedBox(
    constraints: const BoxConstraints(
      minWidth: 70, minHeight: 70,
      maxWidth: 150, maxHeight: 150,
    ),
    child: Container(color: red, width: 100,
                     height: 100),
  ),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.13: The screen forces the UnconstrainedBox to be exactly the same size as the screen. However, the UnconstrainedBox lets its child Container be any size it wants.

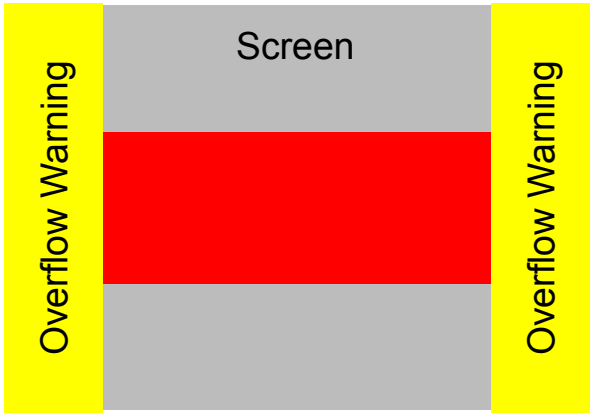



```
UnconstrainedBox(
  child: Container(
           color: red,
           width: 20,
           height: 50),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.14: The Container is 4000 pixels wide and is too big to fit in the UnconstrainedBox, so the UnconstrainedBox displays the much dreaded "overflow warning".

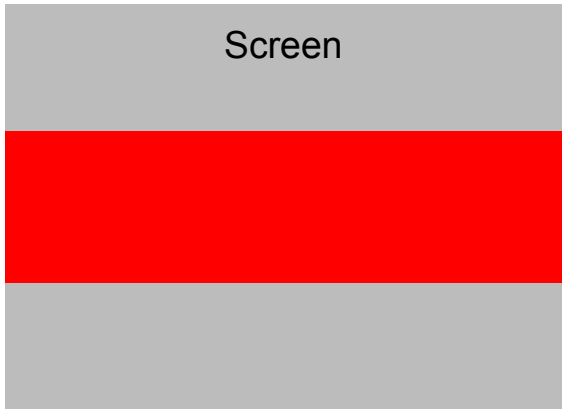



```
UnconstrainedBox(
  child: Container(
           color: red,
           width: 4000,
           height: 50),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.15: The Container has 4000 pixels of width, and is too big to fit in the OverflowBox, but the OverflowBox simply shows as much as it can, with no warnings given.

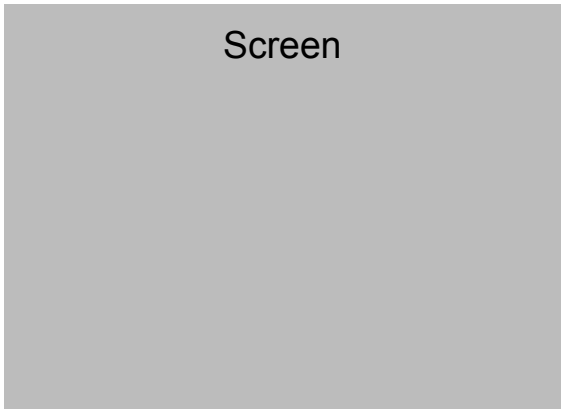



```
OverflowBox(
  minWidth: 0,
  minHeight: 0,
  maxWidth: double.infinity,
  maxHeight: double.infinity,
  child: Container(color: red, width: 4000,
                   height: 50),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.16: The system can't render infinite sizes, so it throws an error with the following message: BoxConstraints forces an infinite width.

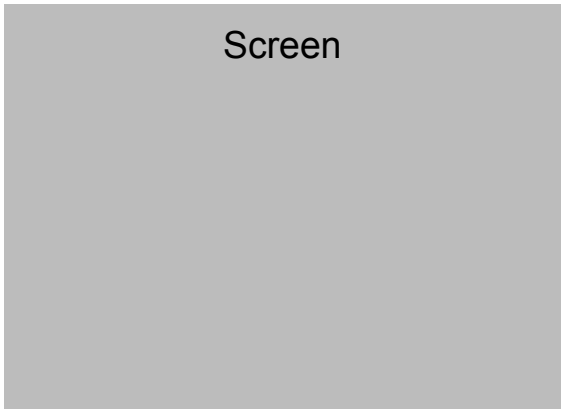



```
UnconstrainedBox(
  child: Container(
          color: Colors.red,
          width: double.infinity,
          height: 100),
)
```

Architectural - Reactive UI

2. Declarative UI

E.g.17: The system can't render infinite sizes, so it throws an error with the following message: BoxConstraints forces an infinite width.




```
UnconstrainedBox(
  child: Container(
          color: Colors.red,
          width: double.infinity,
          height: 100),
)
```

Architectural - Reactive UI

2. Declarative UI

“Constraints go down. Sizes go up. Parent sets position.”

- A widget can decide its own size only within the constraints given to it by its parent.
- Widget's parent who decides the position of the widget.
- It's impossible to precisely define the size and position of any widget without taking into consideration the parent-child tree as a whole.
- If a child wants a different size from its parent and the parent doesn't have enough information to align it, then the child's size might be ignored.

Architectural - State Management

State refers to the condition or data of a system at a particular moment.

It's a snapshot of the system's variables, properties, and other data that define its behavior.

In Flutter, state determines the dynamic behaviour of your app. Two main types of state:

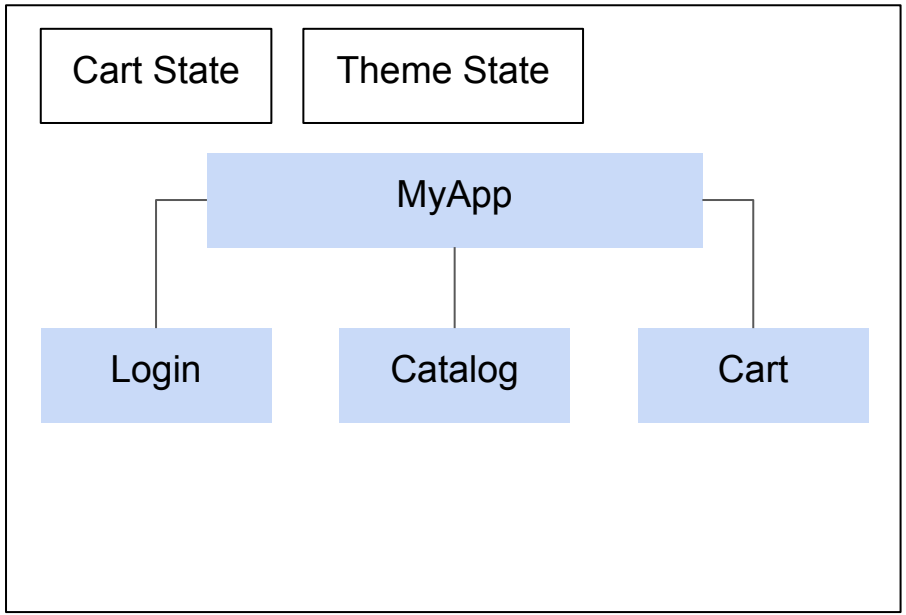
1. App State
2. Widget State

Architectural - State Management

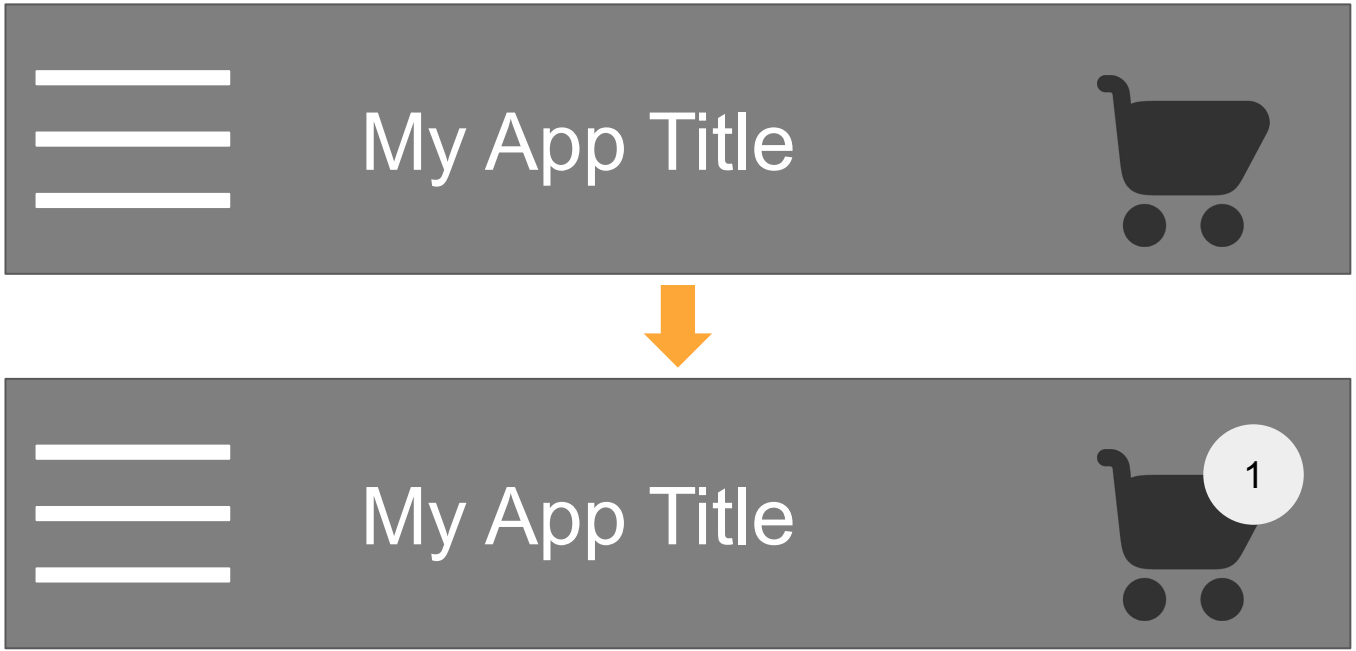
1. App State

Global state that affects the entire app.

Managed using libraries like `Provider`, `Riverpod`, or `BLoC`.



Architectural - State Management

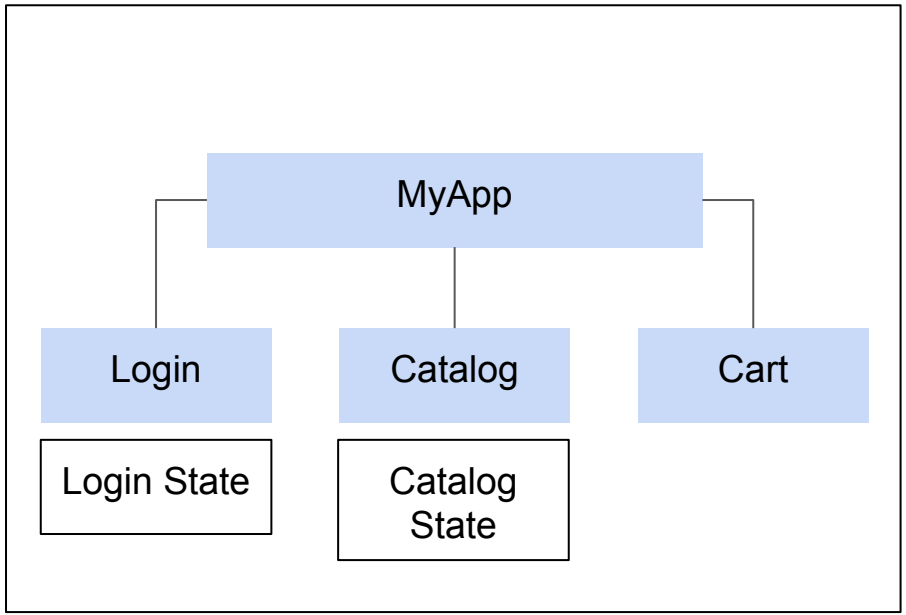


Architectural - State Management

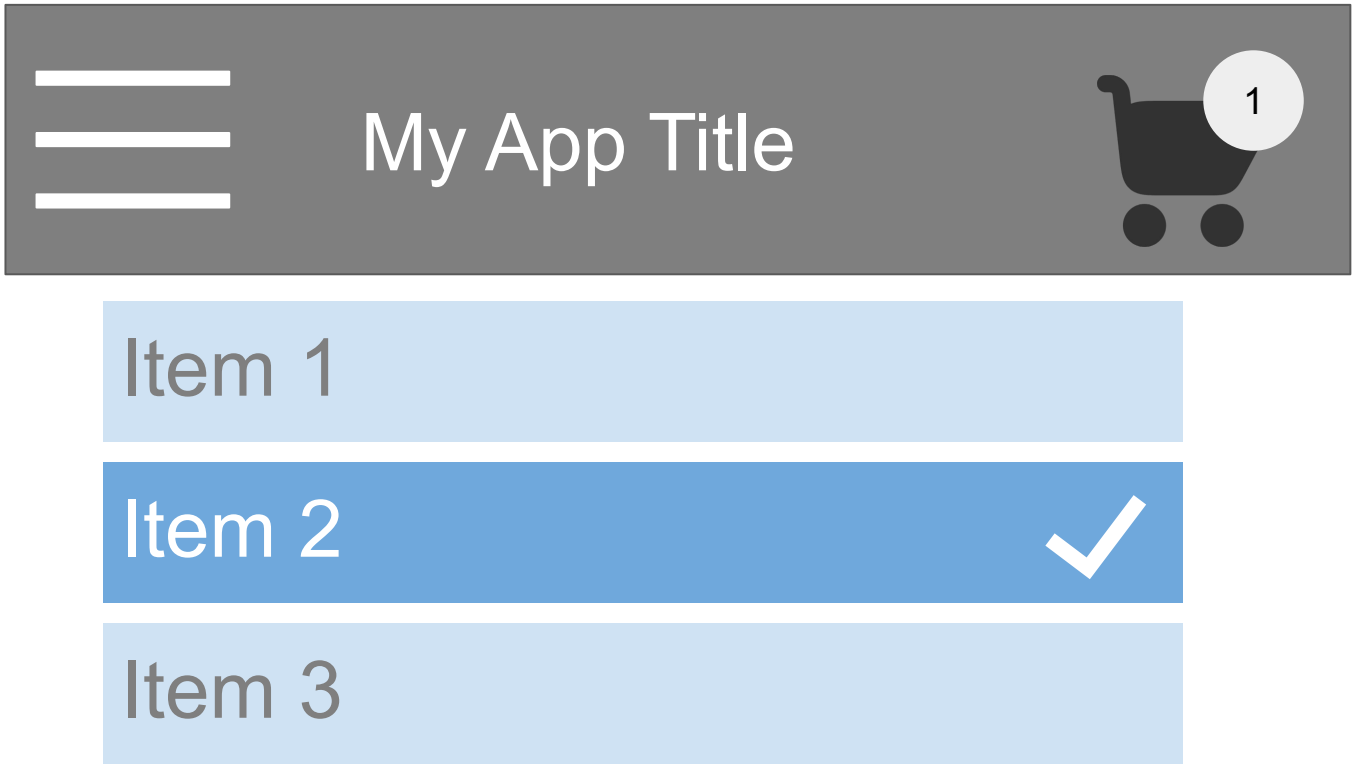
2. Widget State

Local state specific to a particular widget.

Managed using `StatefulWidget` and `setState`



Architectural - State Management



React Framework

Flutter is a reactive, declaration UI framework.

Widgets are the building blocks of a Flutter app's UI.

Developer provides:

1. A mapping from application state to UI.
2. The framework takes on the task of updating the UI at runtime when the application state changes.

React Framework

Feature	Traditional UI	Reactive UI
Programming Paradigm	Imperative	Declarative
State Management	Manual	Framework-assisted
UI Updates	Manual	Automatic
Performance	Can be less efficient	More efficient

```
class MainActivity : AppCompatActivity() {
    private lateinit var textView: TextView

    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContentView(R.layout.activity_main)

        textView = findViewById(R.id.textView)

        // Button click handler
        button.setOnClickListener {
            val newText = "Hello, World!"
            textView.text = newText
        }
    }
}
```

Declare UI state

Layout of UI

Linking UI to state

Update state and UI

```
class _MyHomePageState extends State<MyHomePage> {
  String _text = '';

  void _toggleText() {
    setState(() { _text = 'Hello, World!''});
  }

  @override
  Widget build(BuildContext context) {
    return Scaffold(
      appBar: AppBar(
        title: const Text('My App'),
      ),
      body: Center(
        child: Column(
          mainAxisAlignment: MainAxisAlignment.center,
          children: [
              Text(  _text, style: const TextStyle(fontSize: 24),
            ),
            ElevatedButton( onPressed: _toggleText,
              child: const Text('Click Me'),
        ...
}
```

Declare UI state

Update state

Layout of UI

React Framework - Widget

Widgets form a hierarchy based on composition.

Each widget nests inside its parent and can receive context from the parent.

This structure carries all the way up to the root widget (the container that hosts the Flutter app, typically MaterialApp or CupertinoApp).

Note: We will discuss widgets in more detail…

Conclusion

-Mobile operating systems and ecosystem

-Development tools

-Mobile devices

-Mobile application software architecture and framework

Find Out More

Flutter architectural overview,
https://docs.flutter.dev/resources/architectural-overview

Chapter 1.2

Introduction to Dart

Introduction

Dart is an open-source, general-purpose programming language developed by Google

Applications:

- Web development
- Mobile development
- Server-side development

Introduction

Key features:

1. Object-oriented
2. Strongly typed
3. Fast and efficient
4. Modern features

Structure

```
// This is a comment
void main() {
   // This is the main function, program starts here
   print("Hello, world!");
}
```

Variable

Data Types

Numbers:

- `int:` Stores whole numbers (integers).
- `double:` Stores numbers with decimals (floating-point numbers).
- `num:` Supertype of both int and double.

Variable

Declaration Keywords

```
// Declares an integer variable, needs initialization before use
int age;
// Declares a double variable
double pi = 3.14159;
// Declare multiple variables of the same type in a single line
int x = 5, y = 10;
```

Variable

Declaration Keywords

```
// Nullable double, initially null

double? area = null;
```

Variable

Data Types

Text:

- `String:` Stores sequences of characters (text).

Variable

Declaration Keywords

```
// String type with initial value

String name = "Alice";

// var infers String based on assignment

var greeting = "Hello";

// var infers String based on assignment

var flybyObjects = ['Jupiter', 'Saturn', 'Uranus', 'Neptune'];
```

Variable

Data Types

Logical:

- `bool:` Stores true or false values (Boolean).

Variable

Declaration Keywords

```
bool isLoggedIn = true;

bool isNightTime = false;
```

Basic Structures

```
if (year >= 2001) {
  print('21st century');
} else if (year >= 1901) {
  print('20th century');
}
```

Basic Structures

```
for (int month = 1; month <= 12; month++) {
  print(month);
}

for (final object in flybyObjects) {
  print(object);
}
```

Basic Structures

```
while (year < 2016) {
  year += 1;
}
```

Functions

```
bool isNoble(int atomicNumber) {
  return _nobleGases[atomicNumber] != null;
}
```

Functions

```
int fibonacci(int n) {
  if (n == 0 || n == 1) return n;
  return fibonacci(n - 1) + fibonacci(n - 2);
}

var result = fibonacci(20);
```

Functions

```
bool isNoble(int atomicNumber) => _nobleGases[atomicNumber] !=

null;


flybyObjects.where((name) =>name.contains('turn')).forEach(print);
```

Find Out More

Intro to Dart